SHENZHEN SILICONTRA TECHNOLOGY CO.,LTD.

PAN3029ZTR4-GC

433MHz Wireless Transceiver Module User Specification

(V1.0)

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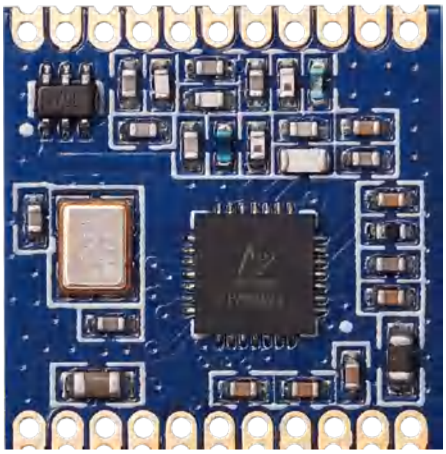
Document Revision History

Version	Change Date	Change Description
V1.0	August 5, 2024	Initial release

1. Introduction



Top side

Bottom side

Module view

1.1 Module Overview

PAN3029ZTR4-GC is a module independently developed and designed by our company based on the RF chip PAN3029 of Panqi Micro, a domestic original manufacturer in my country.

The biggest highlight of the PAN3029ZTR4-GC module is that the transmission power reaches +20dBm and the receiving sensitivity reaches -143dbm. Coupled with its advanced ChirploT™ modulation and demodulation technology, the communication distance has been greatly improved. It has a very strong position in some short-distance IoT wireless communication fields.

The PAN3029ZTR4-GC module has the characteristics of small size, low power consumption, long transmission distance, and strong anti-interference ability. It can be widely used in various wireless communication fields of the Internet of Things.

1.2 Module Features

- Support 433MHz frequency band, strong penetration
- Maximum transmission power 20dBm, power software adjustable, receiving current 4.1mA@DCDC mode
- SPI communication interface, can be directly connected to various microcontrollers with wide voltage working range 1.8~3.6V
- Industrial standard design, support long-term use at -40 ~+85°C Ultra-small size, only 15x15mm
- Stamp hole design, convenient for mass production

1.3 Application Scenarios

- Home security alarm
- Smart home
- Smart parking systems
- Agricultural automation solutions
- Wireless industrial remote control, industrial sensors and other products
- Healthcare
- Building automatic meter reading system
- Automotive industry applications

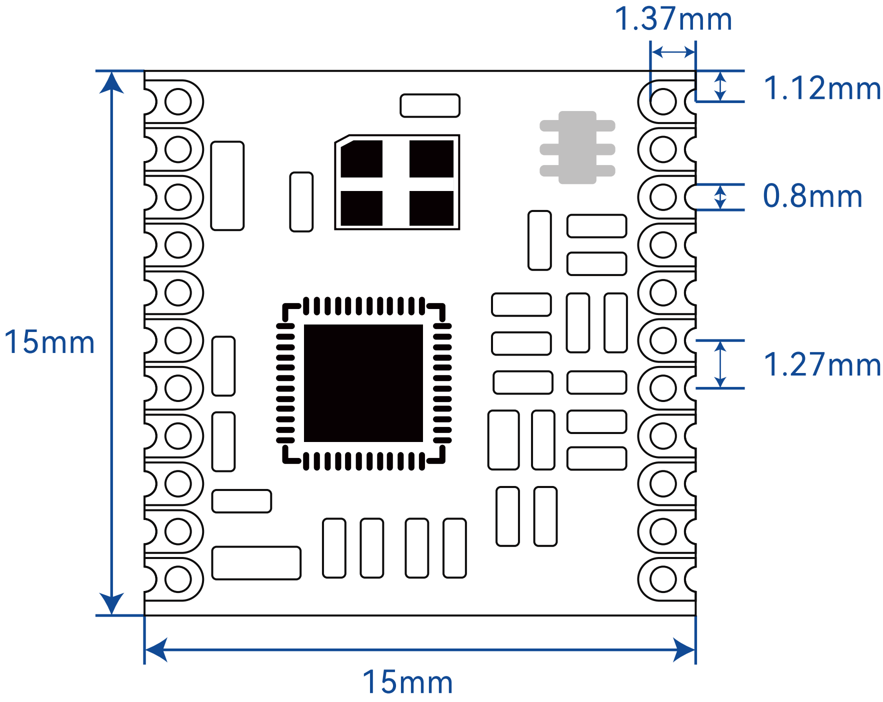
2. Module parameters

2.1 Basic electrical parameters

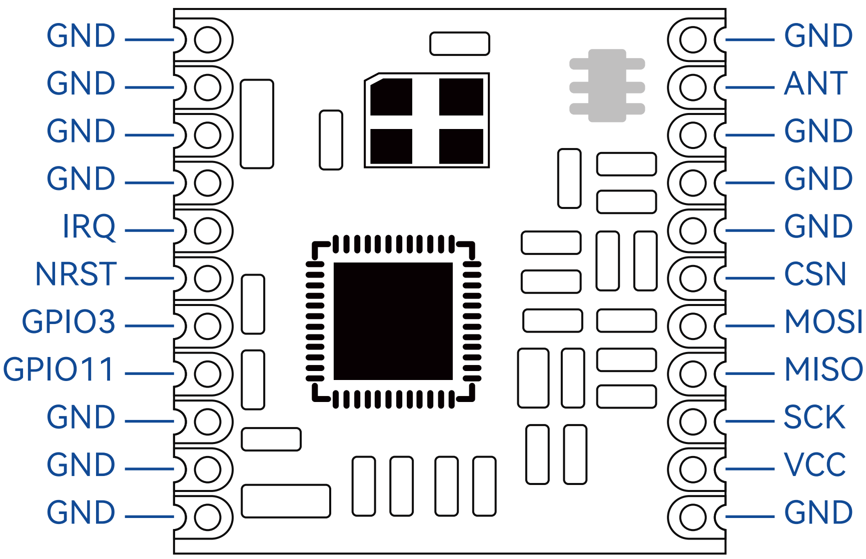
Parameter	Specification		Note
Operating voltage	1.8 ~ 3.6V		3.3V typ.
Operating temperature	-40 ~ 85°C		
Operating frequency	408 ~ 550MHz		433MHz typ., software configurable
Power consumption	Transmit mode	96mA@20dBm	Maximum transmit power
	Receive mode	8mA@LDO / 4.1mA@DC-DC	
	Sleep mode	260nA	
Transmit power	20dBm		Maximum value, user programmable
Receiving sensitivity	-143dBm		@62.5KHz
Modulation	ChirpIoT™		
Modulation parameters	BW：62.5KHZ~500KHZ；SF：5~12；CR：4/5, 4/6, 4/7, 4/8		
Mounting	Stamp Holes		
Communication interface	SPI		
Dimensions	15*15mm		
Antenna Type	Stamp hole external antenna		Equivalent impedance 50Ω

3. Module Description

3.1 Module Dimensions



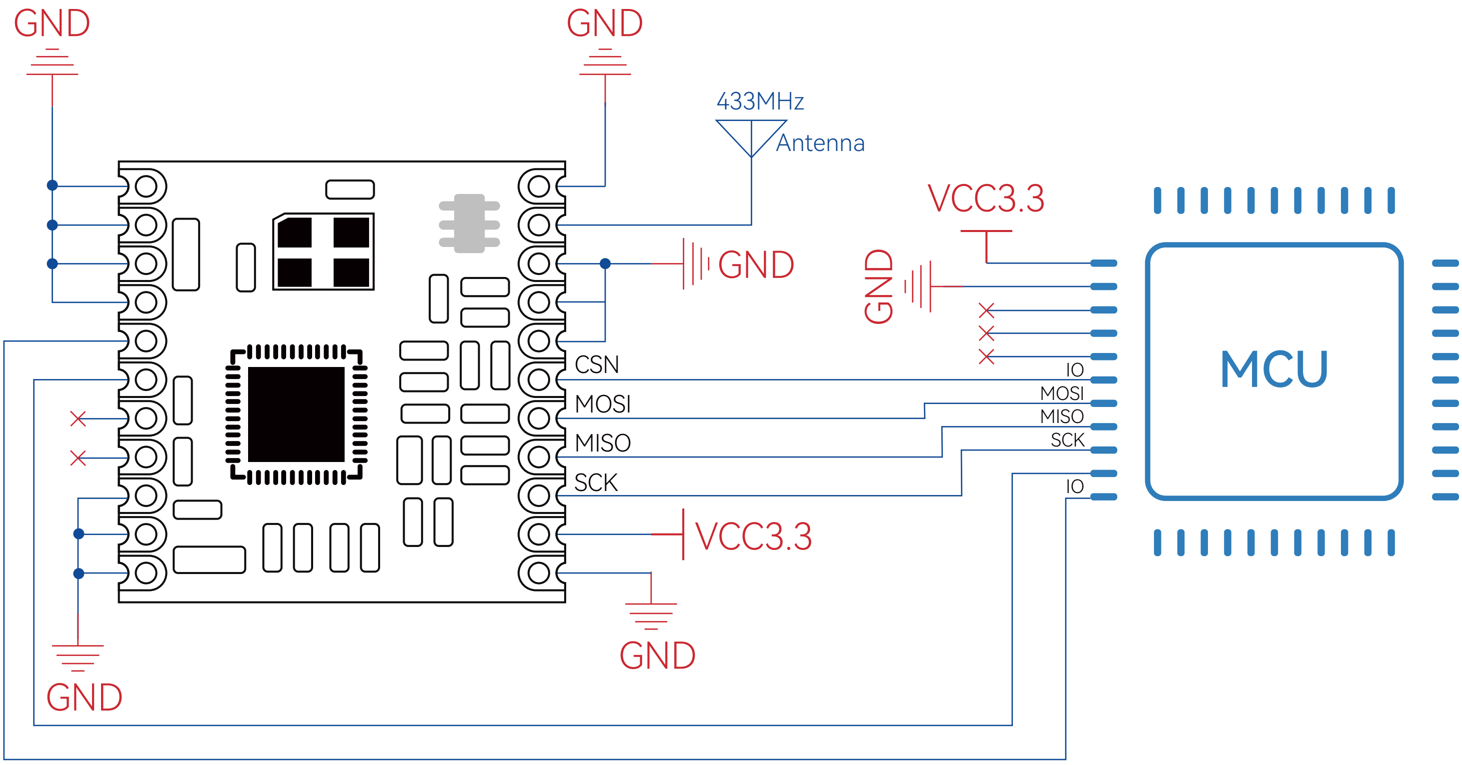
3.2 Module pin definition diagram



3.3 Pin Function Description

Pin No.	Pin name	Functional Description
1	GND	Ground
2	GND	Ground
3	GND	Ground
4	GND	Ground
5	IRQ	Interrupt signal
6	NRST	Reset
7	GPIO3	Digital IO
8	GPIO11	Digital IO or channel status indication signal
9	GND	Ground
10	GND	Ground
11	GND	Ground
12	GND	Ground
13	VCC	Module power pin, typical supply voltage 3.3V
14	SCK	SPI bus clock pin, normally high level
15	MISO	SPI bus slave data output pin
16	MOSI	SPI bus slave data input pin
17	CSN	SPI chip select pin, low level active
18	GND	Ground
19	GND	Ground
20	GND	Ground
21	ANT	Antenna interface, equivalent impedance close to 50Ω
22	GND	Ground

3.4 Module connection diagram



4. Additional Notes

1. It is recommended to use a linear regulated power supply (LDO) to power the module. The power ripple coefficient should be as small as possible. The module needs to be reliably grounded. Please pay attention to the correct connection of the positive and negative poles of the power supply. Reverse connection may cause permanent damage to the module.

2. No other metal objects should be placed near the module antenna, otherwise it will seriously affect the communication distance.

5. Antenna Selection

5.1 Antenna Use Precautions

- The antenna installation structure has a great impact on the performance of the module. For better results, the antenna needs to be exposed, preferably vertically upward. When the module is installed inside the housing, you can use a high-quality antenna extension cable to extend the antenna to the outside of the housing; if the product is not allowed to be exposed, you need to match the spring antenna or FPC antenna.
- If the antenna is installed inside the metal shell, the transmission distance will be greatly reduced.
- If you choose a suction cup antenna, straighten the lead as much as possible and attach the suction cup base to the metal object as much as possible.

433MHz弹簧天线

433MHzFPC天线

433MHz棒状天线

433MHz吸盘天线

6. Hardware Design

- It is recommended to use a DC regulated power supply to power the module. The power supply ripple coefficient should be as small as possible, and the module needs to be reliably grounded.
- Please pay attention to the correct connection of the positive and negative poles of the power supply. Reverse connection may cause permanent damage to the module.
- Please check the power supply and make sure it is within the recommended voltage range. If it exceeds the maximum value, the module may be permanently damaged.
- Please check the stability of the power supply. The voltage cannot fluctuate greatly and frequently.
- When designing a power supply circuit for a module, it is often recommended to retain a margin of more than 30%, which is conducive to the long-term and stable operation of the entire machine.
- The module should be kept as far away as possible from parts with large electromagnetic interference, such as power supplies, transformers, and high-frequency wiring.
- High-frequency digital routing, high-frequency analog routing, and power routing must avoid the bottom of the module. If it is necessary to pass under the module, assuming that the module is soldered on the Top Layer, ground copper is laid on the Top Layer of the module contact part (all copper is laid and well grounded), and it must be close to the digital part of the module and routed on the Bottom Layer.
- Assuming the module is soldered or placed on the Top Layer, it is also wrong to randomly route the wires on the Bottom Layer or other layers, which will affect the module's spurious signal and receiving sensitivity to varying degrees.
- If there are devices with large electromagnetic interference around the module, it will also greatly affect the performance of the module. It is recommended to keep them away from the module according to the intensity of the interference. If possible, appropriate isolation and shielding can be performed.
- If there are traces with large electromagnetic interference around the module (high-frequency digital, high-frequency analog, power traces), it will also greatly affect the performance of the module. According to the intensity of the interference, it is recommended to stay away from the module appropriately. If the situation permits, appropriate isolation and shielding can be performed.

7. The transmission distance is not ideal

- When there are obstacles or obstructions in straight-line communication, the communication distance will be attenuated accordingly.
- Temperature, humidity, and co-channel interference will increase the communication packet loss rate.
- The ground absorbs and reflects radio waves, and the test effect is poor near the ground.
- There are metal objects near the antenna, or it is placed in a metal shell, the signal attenuation will be very serious. The air rate is set too high (the higher the air rate, the closer the distance).
- The power supply voltage at room temperature is lower than the recommended value. The lower the voltage, the lower the power.
- The antenna is poorly matched with the module or the antenna itself has quality problems.

8. The module is easily damaged

- Please check the power supply to ensure that it is within the recommended power supply voltage. If it exceeds the maximum value, the module will be permanently damaged.
- Please check the stability of the power supply. The voltage cannot fluctuate.
- Please ensure anti-static operation during installation and use. High-frequency components are electrostatically sensitive.
- Please ensure that the humidity during installation and use is not too high. Some components are humidity-sensitive devices.
- If there is no special requirement for the product, it is not recommended to use it at too high or too low temperature.

9. The bit error rate is too high

- If there is interference from the same frequency signal nearby, stay away from the interference source or change the frequency or channel to avoid interference.
- An unsatisfactory power supply may also cause garbled code. Ensure the reliability of the power supply.
- If the extension line or feeder is of poor quality or too long, it will also cause a high bit error rate.